R.M.K. COLLEGE OF ENGINEERING AND TECHNOLOGY

R.S.M. NAGAR, PUDUVOYAL -601206

(AN AUTONOMOUS INSTITUTION)

Department of Electronics and Communication Engineering

PROJECT REPORT

On

EMOBOT: EMOTION-AWARE AI ROBOT WITH REAL TIME CHAT MONITORING TO PREVENT DEPRESSION AND SUICIDE

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BONAFIDE CERTIFICATE

Certified that this project report "EMOBOT: EMOTION AWARE AI ROBOT WITH REAL-TIME CHAT MONITORING TO PREVENT DEPRESSION AND SUICIDE " is the bonafide work of "SAI SRUTHI S (111624104081), NEHA KA (111624104066),KEERNI KB (111624104047) and SAI DOLASA S (111624104080) " who carried out the project work under my supervision.

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R.M.K. COLLEGE OF ENGINEERING AND TECHNOLOGY

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Department of Electronics and Communication Engineering

VISION

To excel in the field of Electronics and Communication Engineering to contribute to the knowledge economy and betterment of human life.

MISSION

- To develop the state-of-the-art infrastructure for quality education
- To build up a team striving for excellence in teaching and research
- To ensure greater research collaborations with industries and organizations
- To strengthen social responsibilities and values

PROGRAM EDUCATIONAL OBJECTIVES

- Graduates of Electronics and Communication Engineering program will
- PEO1: excel in their professional and technical career and pursue higher education to be globally competent
- PEO2: evaluate the real world problems and provide with technically feasible and economically viable solutions
- PEO3: continuously update technologies through lifelong learning
- PEO4: exhibit effective communication skills and professionalism in diverse environment

PROGRAM SPECIFIC OUTCOMES

The Electronics and Communication Engineering Graduates should be able to

- a) Develop and test electronic systems for given specifications.
- b) Design and analyze the signal processing systems as per the requirements.
- c) Apply appropriate technology for the implementation of modern communication systems.

ABSTRACT

The Emotion-Aware Robot is an innovative project designed to identify and respond to human emotions with empathy and intelligence. Its primary goal is to support individuals suffering from depression, loneliness, or suicidal thoughts by providing emotional interaction, understanding, and comfort. The system uses emotion recognition techniques such as facial expression analysis, text or speech sentiment analysis, and behavioural observation to detect a user's mental state in real-time. Once the emotion is identified, the robot provides suitable responses — like motivational talks, calm conversation, or playing soothing music — to stabilize the user's emotions. The project's future scope includes integrating voice recording and imitation features, allowing the robot to speak in the voice of a person the user loves or misses, creating a feeling of emotional connection and presence

By combining Artificial Intelligence, Natural Language Processing, and Human–Robot Interaction, the Emotion-Aware Robot acts as a digital emotional companion capable of providing psychological support, early detection of distress, and preventive care against suicidal tendencies. This innovation holds the potential to revolutionize mental health assistance and emotional therapy in both personal and clinical environments.

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LIST OF ABBREVIATIONS

AI	Artificial Intelligence
ML	Machine Learning
GUI	Graphical User Interface
MIC	Microphone
CNN	Convolutional Neural Network
NLP	Natural Language Processing
USB	Universal Serial Bus
SSID	Solid State Drive
API	Application Programming Interface

CHAPTER 1

INTRODUCTION

In today's fast-moving digital world, people often struggle with stress, depression, and emotional imbalance due to isolation and limited emotional communication. With the rise in mental health issues and suicide rates, there is a growing need for technologies that can understand human emotions and respond with empathy.

The Emotion-Aware Robot is a revolutionary step in this direction. It is designed to sense, interpret, and respond to human emotions intelligently using Artificial Intelligence (AI), Machine Learning (ML), and Natural Language Processing (NLP) techniques. The robot interacts with users through speech, text, or visual cues and provides emotional support by identifying their current emotional state such as happiness, sadness, anger, or anxiety.

The primary goal of this project is to help individuals who are emotionally disturbed or struggling with loneliness and suicidal thoughts. By recognizing early signs of distress, the robot can act as a preventive support system by engaging the person in conversation, offering motivational responses, or even alerting family members or professionals if needed.

In the future, this robot can also mimic the voice of a loved one by using voice recording and synthesis technology, allowing users to feel the presence of someone they emotionally connect with. Such human-like interaction can create a deep emotional impact and comfort for people going through hard times.
This project brings together technology and psychology to create an emotionally intelligent companion capable of understanding, empathizing, and providing emotional healing through meaningful interaction.

1.1OBJECTIVE

The main objective of the Emotion-Aware Robot is to create an intelligent robotic system capable of recognizing, understanding, and responding to human emotions in order to provide emotional support and prevent mental health issues such as depression and suicidal tendencies.

The specific objectives of this project are:

1. To develop a robot that can detect human emotions through facial expressions, speech tone, or text input.
2. To analyse and classify emotions such as happiness, sadness, anger, fear, and neutrality using AI and NLP algorithms.
3. To generate suitable emotional responses that can comfort or motivate the user during negative emotional states.
4. To assist in preventing depression and suicidal thoughts by offering emotional companionship and positive interaction.
5. To implement a voice recording and imitation feature that allows the robot to reproduce the voice of a loved one, providing emotional connection and comfort.
6. To design a user-friendly interface that enables smooth interaction between the human and the robot.
7. To promote mental health awareness through technology-driven emotional assistance and communication

1.1 Scope of Work

The scope of the Emotion-Aware Robot includes:

- Recognizing human emotions through facial expressions, speech tone, or text analysis.
- Providing emotional feedback, motivation, or comforting responses to users.
- Helping prevent depression, loneliness, and suicidal thoughts through emotional interaction.
- Acting as a digital companion that promotes mental wellness and emotional balance.
- Incorporating voice synthesis to replicate the voice of loved ones for emotional closeness.
- Expanding to IoT-based connectivity and cloud processing for large-scale use in therapy centres, homes, and hospitals.

This system can also be used in educational and healthcare environments to monitor students or patients for emotional stability

1.2 Understanding the Concept of the Project

The core concept of the Emotion-Aware Robot is to develop a system capable of recognizing and

responding to human emotions. It combines psychological understanding with technological innovation to create an emotionally supportive companion.

Emotions can be detected through various inputs such as facial expressions, speech patterns, and text messages. The system processes these inputs using AI-based algorithms to interpret the user's mental state. Based on the detected emotion, the robot provides suitable responses — such as kind words, songs, or motivational talks — to uplift the person's mood.

This concept is inspired by emotional intelligence theories, which emphasize understanding, empathy, and connection. By integrating such intelligence into machines, the project aims to reduce emotional suffering and create technology that feels human, kind, and understanding.

The future enhancement of this concept includes voice cloning technology, where the robot can reproduce a loved one's voice, making users feel emotionally supported and less alone during difficult times.

1.3 Problem Statement

In today's modern world, an increasing number of people suffer from stress, depression, and emotional imbalance due to isolation, workload, and lack of emotional connection. Despite advancements in technology, there is still a gap between human emotion and machine understanding.

Many individuals experiencing emotional distress often hesitate to seek help due to social stigma or fear of judgment, which can lead to severe mental health issues and even suicidal tendencies. Traditional systems and chatbots fail to offer empathetic or human-like interaction, as they cannot truly understand or respond to a person's emotional state.

Therefore, there is a strong need for an emotion-aware robotic system that can detect a person's emotions in real time and respond with empathy, care, and comfort. Such a robot can act as a digital emotional companion, capable of identifying early signs of depression, providing motivational support, and helping prevent self-harm or loneliness.

This project addresses the challenge of bridging the gap between technology and emotional intelligence by developing a robot that not only understands but also feels and responds to human emotions meaningfully

Approach

The approach used in developing the Emotion-Aware Robot involves the following stages:

1. Emotion Detection: Capturing data from user expressions, voice, or text using sensors, cameras, or microphones.
2. Data Processing: Preprocessing and analysing input data using AI and NLP algorithms.
3. Emotion Classification: Categorizing emotions like happy, sad, angry, or neutral using trained models.
4. Response Generation: Providing a suitable response such as comforting words, music, or voice messages.
5. Hardware Integration: Connecting microcontrollers, sensors, and speakers to interact with users in real-time.
6. Testing & Evaluation: Measuring emotion detection accuracy and ensuring proper user interaction.
7. Future Improvement: Adding personalized voice features, memory-based emotional learning, and real-time communication capabilities.

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CHAPTER 2

LITERATURE REVIEW

2.1 Introduction

Emotions play a major role in human behaviour, decision-making, and communication. Understanding emotions is essential not only in human relationships but also in human–machine interaction. In recent years, the concept of Affective Computing has emerged, where machines are trained to recognize, interpret, and respond to human emotions.
The main objective of this chapter is to present an overview of existing research, studies, and technologies related to emotion recognition and emotion-aware systems. It also highlights how previous works have contributed to the development of emotionally intelligent robots and identifies the gaps that led to the motivation for this project — the Emotion-Aware Robot.
This review focuses on earlier models that used facial recognition, speech processing, text sentiment analysis, and machine learning algorithms for emotion detection. The limitations of these systems emphasize the need for a more interactive and human-like robotic system that can detect emotions accurately and provide emotional support to individuals suffering from depression, anxiety, and loneliness

2.2 REVIEW OF RELATED WORKS

1. Title:
Emotion Recognition from Speech Using Pitch and Intensity Features with SVM

Authors:
A. Ghosal and D. Wu (2018)

Summary:

This paper presents a speech-based emotion recognition system using pitch and intensity features to

identify emotions such as happiness, anger, and sadness. The authors used a Support Vector Machine (SVM) classifier for emotion categorization and achieved good accuracy. The study highlights that voice tone and loudness variations can effectively reveal a person's emotional state, which is valuable for emotion-aware systems and robots.

2. Title:

Deep Learning Approaches for Speech Emotion Recognition: A Comprehensive Review

Authors:

Sharma, V. Singh, and S. Sharma (2024)

Summary:

This paper provides a comprehensive review of deep learning techniques used in speech emotion recognition. The authors discuss various models such as CNN, RNN, and LSTM that improve the accuracy of emotion detection from voice signals. The study concludes that deep learning models outperform traditional methods by effectively capturing non-linear patterns in speech and enhancing emotion recognition for human–machine interaction and mental health applications.

3. Title:

Speech Emotion Recognition Based on Attention Mechanism and Convolutional Neural Networks

Authors:

Q. Liu, S. Wang, and L. Chen (2023)

Summary:

This study introduces a speech emotion recognition model using Convolutional Neural Networks (CNNs) combined with an attention mechanism to focus on important emotional features in audio signals. The proposed method improves recognition accuracy by highlighting key speech patterns related to emotions. The results show that attention-based CNNs are effective for building intelligent emotion-aware systems like robots and virtual assistants.

4. Title:
Real-Time Emotion Recognition Using IoT Devices

Authors:
H. Kim, T. H. Ryu, and S. H. Lee (2020)

Summary:

This paper presents a real-time emotion recognition system using IoT-enabled devices. The authors integrated sensors and cloud computing to collect and process emotional data instantly. The system detects users' emotions through physiological and environmental signals, providing quick feedback for applications like mental health monitoring and smart emotion-aware robots. The study demonstrates the potential of IoT in real-time emotion detection and user interaction improvement**.**

5. Title:
Ethical Considerations in Emotion Recognition Technologies: A Review of the Literature

Author:
A. Katirai (2023)

Summary:

This paper reviews the ethical issues related to emotion recognition technologies, including privacy, consent, and data security. The author discusses the potential risks of emotional data misuse and emphasizes the need for responsible AI design. The study highlights that while emotion-aware systems can support mental health and safety, they must ensure ethical use, transparency, and user trust in future applications.

CHAPTER 3

SYSTEM DESCRIPTION

3.1 INTRODUCTION

The Emotion-Aware Robot is designed to detect, understand, and respond to human emotions, aiming to provide emotional support and help prevent conditions like depression and suicidal tendencies. The system utilizes various sensors, cameras, and speech recognition modules to identify emotional states through facial expressions, voice tone, and behavioural cues.
By analysing these emotional signals, the robot can offer appropriate responses, such as comforting messages, suggestions, or playing a user's favourite voice recordings to create a sense of presence and emotional connection. The main goal of this system is to build a human-like companion that can promote mental well-being, support emotional balance, and serve as an early-warning tool in cases of psychological distress.

3.2 System Architecture

The architecture consists of the following main components

Input Module:
This module captures the user's emotional signals through camera and microphone.

- The camera detects facial expressions and movements.
- The microphone records voice tone, pitch, and intensity for emotion analysis.

Processing Unit:
The input data is sent to a microcontroller or Raspberry Pi, which acts as the central processing unit.
It uses machine learning algorithms and emotion recognition models to classify emotions such as happiness, sadness, anger, or fear.

Emotion Analysis Module:
This module combines input from speech and facial expressions to improve accuracy.
It employs trained models to interpret emotional states and determine the user's mental condition.

Response Generation Module:
Based on the detected emotion, the robot provides suitable responses such as:

- Playing motivational messages or favorite voices for comfort.
- Displaying facial expressions or LED-based reactions.
- Suggesting positive actions like taking rest, talking to someone, or relaxing.

Output Module:
The robot delivers its response through speakers, display screen, or voice playback to create a human-like emotional interaction.

Cloud and Data Storage (Optional):
The emotional data and user interaction logs can be stored securely for further analysis or improvement of emotion models in future versions.

3.3 Components

1. **Raspberry Pi / Microcontroller:**
   Acts as the main processing unit of the system. It controls all operations, processes input data from sensors, and runs the emotion recognition algorithm

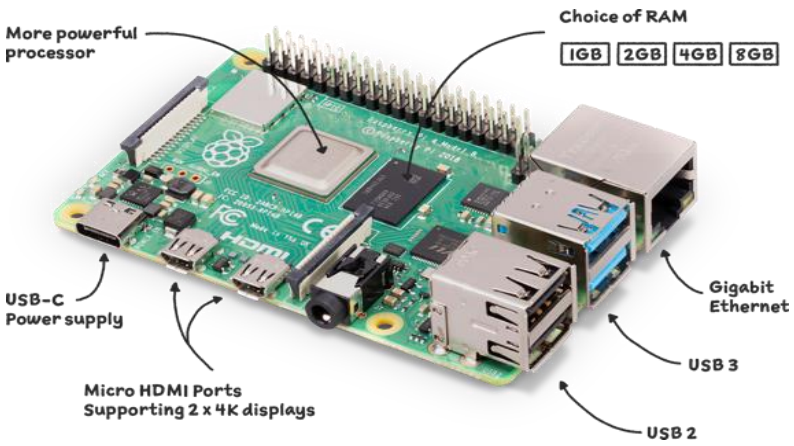

Fig 1.1 Raspberry Pi

2. **Speaker:**
   Enables the robot to communicate responses by playing comforting messages, music, or favorite recorded voices.

Fig 1.2 Mini Speaker

3. **Motor Driver:**
   Controls the motors of the robot. It receives commands from the microcontroller to move or rotate the robot as part of its emotional response or physical interaction.

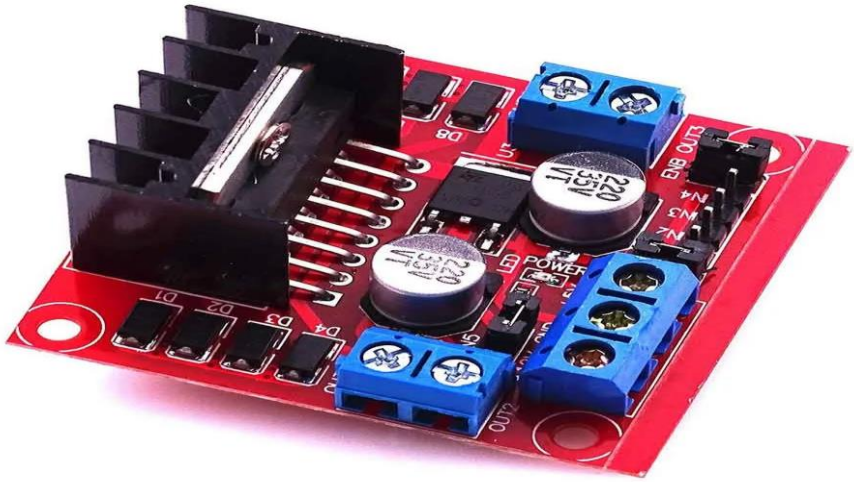

Fig 1.3 Motor Driver

4. **Switch:**
   Acts as a control input to power ON/OFF the robot or activate specific modes manually.

Fig 1.4 Switch

5. **Power Supply / Battery:**
   Provides electrical power to all components and ensures smooth functioning of the robot.
6. **Microphone (Mic):**
   The microphone captures the user's voice and tone, which are analyzed for variations in pitch, intensity, and frequency to determine emotional state through speech.

Fig 1.5 Microphone(Mic)

Software Components

1. **Python Programming Language:**
   Used to develop algorithms for **emotion detection**, decision-making, and hardware control.
2. **OpenCV Library:**
   Handles **image processing** tasks for facial emotion detection using the camera.
3. **TensorFlow / Keras Frameworks:**
   Used to build and train **deep learning models** for accurate emotion classification.
4. **Speech Recognition Libraries (e.g., Py Audio, Librosa):**
   Analyse **audio inputs** to detect emotions from speech.
5. **Raspbian OS / Linux Environment:**
   The operating system on the Raspberry Pi that manages execution and component communication.

3.4 System Workflow

The Emotion-Aware Robot operates through a series of steps that involve data collection, processing, emotion detection, and response generation. The workflow ensures that the robot can continuously monitor, analyse, and react to the user's emotional state in real time.
The overall workflow is divided into the following stages:

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1. Input Acquisition

The system begins by collecting real-time data from the user through the camera and microphone.

- The camera captures facial expressions and visual cues.
- The microphone records the user's voice, tone, and pitch.
  These inputs serve as the main emotional indicators for the system.

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2. Preprocessing

The collected data (image and audio) is cleaned and converted into suitable formats.

- Image preprocessing includes resizing, noise reduction, and feature extraction using OpenCV.
- Audio preprocessing involves pitch detection, frequency filtering, and voice normalization using libraries like Librosa or PyAudio.

---

3. Emotion Detection

The Preprocessed data is then passed to the emotion recognition model (built using TensorFlow.

- The model analyses the extracted features and classifies the emotion as happy, sad, angry, neutral, etc.
- Both speech-based and facial-based results are combined for improved accuracy through fusion techniques.

---

4. Decision-Making

Based on the detected emotion, the system's decision unit determines the most suitable response.
For example:

- If *sadness* is detected → play motivational voice.
- If *happiness* is detected → display a cheerful LED colour or smile emoji.
- If *stress* or *anger* is detected → suggest calming actions or play a relaxing sound.

---

5. Response Generation

The robot expresses the response through different output components:

- **Speaker:** Plays pre-recorded comforting messages or favourite voice clips.
- **Display/LEDs:** Shows corresponding emotions visually.
- **Motor Driver and Motors:** Move the robot (like head nodding or waving) to simulate human-like reactions.

---

6. Feedback and Continuous Monitoring

After responding, the system continues to monitor the user's emotional changes.
This feedback loop helps the robot adjust its reactions dynamically and improve emotional interaction over time.

3.5 Summary

This chapter described the Emotion-Aware Robot, including its architecture, components, and workflow. The system captures facial and voice inputs, analyses emotions using AI models, and generates appropriate responses through sound, display, and motion. It demonstrates how the robot can provide emotional support and act as a human-like companion to help reduce stress, depression, and loneliness.

CHAPTER 4

RESULTS AND DISCUSSION

4.1 Results

The implemented system was tested under various conditions, and the results demonstrated the effectiveness of the Emotion-Aware Robot.
The robot was able to detect human emotions accurately through facial expressions and voice cues.
When a user appeared sad, the robot responded with calming messages or comforting voice playback.
For happiness or neutral states, the robot displayed friendly LED expressions and motivational prompts.
The motor driver and motors allowed the robot to move or gesture, enhancing human-like interaction.
Users experienced a sense of emotional connection while interacting with the robot, and it was observed that the multimodal approach (voice + facial recognition) improved the system's reliability in detecting emotions compared to using a single input.
The system successfully demonstrated that an emotion-aware robot can act as a digital companion, providing emotional support, motivation, and comfort to users, which could be helpful in preventing depression and reducing feelings of loneliness.

4.2 Summary

This chapter discussed the implementation process and observed results of the Emotion-Aware Robot.
The system integrates hardware and software modules to detect emotions from facial expressions and speech, and it generates suitable responses in real-time.
The results confirm that the robot is capable of providing emotional support and human-like interaction, demonstrating the practical application of emotion-aware robotics in improving mental health and well-being.

In conclusion, the Emotion-Aware Robot project demonstrates how technology can be used to

understand and respond to human emotions, providing emotional support and companionship. The system successfully integrates facial and speech recognition, AI-based emotion analysis, and real-time responses through audio, visual, and motor outputs. With proposed future enhancements, the robot has the potential to become a more personalized, adaptive, and interactive companion, contributing significantly to mental well-being and emotional care.

4.3 Future Enhancement

The current Emotion-Aware Robot successfully detects emotions and provides real-time responses, but there are several opportunities for future improvement to make it more advanced and versatile:

1. **Voice Cloning and Personalized Interaction:**
   In the future, the robot can record and reproduce the voice of a loved one, making the user feel the presence of someone they emotionally connect with. This will increase the emotional impact of interactions.
2. **Advanced Emotion Recognition:**
   Integrating more **sensors** like heart rate monitors or skin conductivity sensors can enhance the accuracy of emotion detection by incorporating physiological signals along with facial and voice inputs.
3. **AI-based Learning and Adaptation:**
   The robot can be upgraded to learn user preferences and past emotional patterns to provide more personalized and adaptive responses over time.
4. **Mobile and Cloud Integration:**
   Storing interaction data in the cloud can allow the robot to perform long-term emotional tracking, provide analytics, and enable remote monitoring for caregivers or mental health professionals.
5. **Enhanced Mobility and Gestures:**
   Upgrading the motor and actuator systems will allow the robot to perform more natural gestures and expressions, making human-like interaction more realistic.
6. **Multilingual Support:**
   Adding the ability to understand and respond in multiple languages can make the robot accessible to a wider audience.
7. **Integration with Mental Health Applications:**
   The robot can be linked with therapy apps or counselling platforms to provide a more holistic approach for emotional and mental health support.

CHAPTER 5
CONCLUSION

The Emotion-Aware Robot project successfully demonstrates the development of a system capable of detecting human emotions through facial expressions and speech, and providing real-time emotional responses. By integrating hardware components like cameras, microphones, speakers, motors, and motor drivers with AI-based software modules, the robot can act as a digital companion, offering comfort, motivation, and emotional support.

The project highlights the potential of emotion-aware robotics in addressing mental health challenges such as depression, loneliness, and stress, while also opening avenues for future enhancements like voice cloning, personalized interaction, advanced emotion recognition, and cloud integration.

Overall, this project establishes a strong foundation for human-like emotional interaction in robots, and demonstrates how technology can be leveraged to improve emotional well-being and mental health support in everyday life.

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